

Practical Deployment of a Quantized SNN-Equivalent Intrusion Detector on a Commodity MCU NPU

[Blinded for review]

Abstract

Deploying neural intrusion detection systems on microcontroller-class hardware requires balancing classification accuracy against strict latency, memory, and operator constraints. This paper reports a practical deployment study of a quantized neural IDS classifier on the STM32N6570-DK, a commodity ARM Cortex-M55 MCU with the Neural-ART NPU (600 GOPS INT8). Exploiting the approximate equivalence between single-timestep ($T=1$) SNN inference and INT8 quantized ANN inference, we compile a lightweight MLP to the NPU without neuromorphic hardware. We evaluate on three standard benchmarks — NSL-KDD (5-class), UNSW-NB15 (10-class), and CICIDS2017 (15-class) — with up to 20 random seeds per configuration. INT8 NPU inference completes in 0.29–0.46 ms ($2.7\text{--}4.2\times$ faster than CPU-only execution), with a Flash footprint under 140 KB. A 24-configuration quantization ablation on NSL-KDD shows accuracy deviations below 1 percentage point (UNSW-NB15 is less stable), and layer-wise analysis confirms 99% FP32/INT8 prediction agreement on the NSL-KDD benchmark. We characterize the accuracy–complexity Pareto frontier across four deployable model sizes and document that the `Floor` operator required by QCFS activations forces CPU fallback with 17.6% latency overhead. Our evaluation is bounded to classifier inference on pre-extracted features; power consumption is estimated, not measured. The results provide practical deployment evidence and operator-level insight for neural IDS on commodity MCU NPUs.

Keywords: intrusion detection system · spiking neural network · ANN-SNN conversion · quantization · edge AI · neural processing unit

1 Introduction

Deep learning has transformed network intrusion detection, but deploying these models at the network edge remains difficult. Edge IDS must operate under tight power and latency budgets, and the dominant approach today still relies on cloud offloading or high-end GPUs, neither of which fits IoT or telecom CPE (customer premises equipment) scenarios where per-device cost must stay below \$10.

Spiking neural networks offer a different compute paradigm. Because SNNs process information as discrete

spikes rather than dense floating-point activations, they can be extremely energy-efficient when run on neuromorphic hardware [18]. The catch is that neuromorphic chips are not commodity parts. Intel Loihi [5] is a research platform. BrainChip Akida is available commercially but carries a price premium. Neither is a standard MCU that an OEM can drop into a low-cost IoT gateway.

A line of theoretical work changes this picture. Bu et al. [2] introduced QCFS, showing that ANN activations can be mapped to discrete spike counts with minimal loss. Jiang et al. [7] provided the first $T=1$ ANN–SNN conversion framework via SlipReLU. More recently, Bu et al. [3] showed that when an SNN runs for exactly one timestep ($T=1$) with zero initial membrane potential, the threshold-and-fire operation reduces to a ReLU-like clamp, making the forward pass approximately equivalent to an INT8 quantized ANN. The practical implication: any NPU that can execute INT8 matrix multiply and ReLU can run an SNN-equivalent model without neuromorphic hardware.

We put this theory to the test. Using the STM32N6570-DK development board (ARM Cortex-M55 at 800 MHz, Neural-ART NPU rated at 600 GOPS INT8), we train MLP classifiers for intrusion detection on three standard benchmarks (NSL-KDD, UNSW-NB15, and CICIDS2017), quantize them to INT8, and deploy on the NPU through ST Edge AI Developer Cloud. The classifier infers in 0.29–0.46 ms ($2.7\text{--}4.2\times$ faster than CPU-only execution) with a memory footprint under 140 KB. We also characterize the accuracy–complexity Pareto frontier across four deployable model sizes (from logistic regression to the full 4-layer MLP), evaluate focal loss as a training variant for class imbalance, and conduct a 24-configuration quantization ablation. We focus on inference latency and memory footprint of the classifier stage on pre-extracted flow-level features; power estimates derived from ST’s published application note are noted in Section 5.8 but direct on-board measurement is deferred. A complete end-to-end IDS pipeline (packet capture, flow aggregation, feature extraction) is left to future work.

Our contributions:

- A practical deployment study of quantized neural IDS classifiers on a commodity MCU NPU, covering the full pipeline from PyTorch training through INT8 post-training quantization to Neural-ART NPU execution, evaluated across three standard benchmarks (NSL-KDD, UNSW-NB15, CICIDS2017) with up to 20 random

seeds per configuration.

- We empirically validate that $T=1$ SNN-ANN equivalence holds on commercial NPU silicon, with 99% FP32/INT8 prediction agreement, layer-wise cosine analysis, and stability across 24 quantization configurations.
- We provide an operator-level deployment analysis documenting that QCFS-style `Floor` operations trigger CPU fallback with 17.6% latency overhead, and map the accuracy-complexity-latency Pareto frontier of NPU-deployable classifiers from logistic regression through a 4-layer MLP.
- We characterize the deployment constraints honestly: tree ensembles cannot run on the NPU at all, extreme minority-class imbalance limits recall regardless of architecture, and focal loss provides only marginal improvement — establishing the practical boundary of neural IDS on commodity MCU NPUs.

2 Related Work

2.1 SNN-Based Intrusion Detection

Table 1 summarizes prior work on neural-network-based IDS with SNN components or edge hardware deployment. Research on applying SNNs to network intrusion detection has grown rapidly since 2024, though nearly all published results remain in simulation. Wang et al. [23] proposed a convolutional SNN for IDS on NSL-KDD and UNSW-NB15 in software simulation. Prajwalasimha et al. [16] described an event-driven SNN-IDS and tested it in simulation. Mustafa et al. [13] combined recurrent and spiking layers for anomaly detection. Karthik et al. [8] proposed a transformer-spiking hybrid; all run only on GPU.

Two prior works deploy IDS on physical hardware. Zahm et al. [24] used the BrainChip Akida AKD1000 neuromorphic processor, reporting 98.4% accuracy at approximately 1 W. Akida is a purpose-built neuromorphic chip (\$499 dev kit) with native spike processing, not a general-purpose MCU. Ngo et al. [15] deployed a lightweight MLP (1,360 parameters, 11 flow features) on the Analog Devices MAX78000, an AI-specialized MCU with a fixed CNN accelerator. They achieved 98.57% on UNSW-NB15 (binary classification: normal vs. attack) at 18 mW. The MAX78000’s accelerator is hard-wired for convolutional workloads; the STM32N6 Neural-ART NPU is a programmable accelerator that supports arbitrary operator graphs within its INT8 operator set. Our work differs from both in targeting a general-purpose ARM Cortex-M MCU with an attached NPU, using multi-class classification (5-class and 10-class), and validating $T=1$ SNN equivalence.

2.2 ANN-SNN Conversion

The theory of converting a trained ANN into an equivalent SNN has matured over several stages. Bu et al. [2] introduced QCFS, a quantization-aware activation function that maps ANN activations to discrete spike counts. Jiang et al. [7] proposed a unified optimization framework and the first $T=1$ conversion method (SlipReLU). Bu et al. [3] later analyzed the inference-scale complexity of SNNs, establishing channel-wise threshold optimization for $T=1$ conversion. NEXUS [14] recently achieved bit-exact ANN-to-SNN equivalence via modular arithmetic, scaling to LLaMA-2 70B. PASCAL [17] refined multi-timestep conversion with spike accumulation and adaptive layerwise calibration. NeuroFlex [10] showed that co-executing INT8 ANN and SNN layers on the same accelerator can cut the energy-delay product by 57–67%.

To our knowledge, this paper provides the first empirical validation of the $T=1$ equivalence on commercial NPU silicon in the IDS domain.

2.3 Edge AI on MCU NPUs

The STM32N6 is STMicroelectronics’ first MCU with a dedicated neural processing unit [21]. Public deployments span computer vision and audio processing. The NPU’s hardware operator set covers `Conv`, `Gemm`, `Relu`, `Add`, `Clip`, and `MaxPool` in INT8; operators outside this set fall back to the Cortex-M55 CPU [21]. Millar et al. [11] benchmarked micro-NPUs including the Neural-ART; their latency numbers provide a useful reference point. Bruschi et al. [1] deployed SNNs on ARM Cortex-M MCUs and measured energy and latency, but targeted image classification, not IDS, and used software SNN execution without an NPU. To our knowledge, no prior work has deployed an IDS on a general-purpose MCU NPU. The closest prior art (HH-NIDS [15]) used a specialized MCU with a fixed CNN engine. More recently, Chehade et al. [4] deployed a 1D-CNN on the STM32F7 (Cortex-M7, no NPU), requiring 31 ms and 7.86 mJ per inference, and Diab et al. [6] ran LightGBM/CNN on a Raspberry Pi 3B+ (27 ms).

3 Background

3.1 $T=1$ SNN-ANN Equivalence

A Leaky Integrate-and-Fire (LIF) neuron with membrane potential V , weight matrix $\mathbf{W}$, and threshold θ operates as:

$$V[t] = \beta \cdot V[t-1] + \mathbf{W} \cdot \mathbf{x}[t] \quad (1)$$

$$S[t] = \Theta(V[t] - \theta) \quad (2)$$

where Θ is the Heaviside step function and β is the leak factor. At $T=1$ with zero initial state ($V[0] = 0$), the leak term $\beta \cdot V[0]$ vanishes regardless of β , so Eq. 1 simplifies to $V[1] = \mathbf{W} \cdot \mathbf{x}$.

Table 1: Comparison of neural IDS approaches with hardware deployment. “On-chip” indicates deployment on physical hardware. “Task” shows the classification formulation. Accuracy figures are not directly comparable across different task formulations.

Paper	Year	Dataset	Task	Model	Hardware	HW Type	On-chip	Latency	Energy	Acc.
Ngo [15]	2022	UNSW-NB15	binary	MLP	MAX78000	AI-spec. MCU	Yes	—	18 mW*	98.6%
Wang [23]	2024	NSL-KDD	multi	Conv SNN	GPU	GPU	No	—	—	94.7%
Zahm [24]	2024	UNSW-NB15	multi	SNN	Akida AKD1000	Neurom. ASIC	Yes	—	~1 W*	98.4%
Chehade [4]	2025	VPN-nonVPN	multi	1D-CNN	STM32F7	MCU (no NPU)	Yes	31 ms	7.86 mJ	96.6%
Diab [6]	2025	CIC-IDS2017	multi	LightGBM	RPi 3B+	SBC	Yes	27 ms	~6.75 mJ [‡]	95.3%
Prajwal. [16]	2025	NSL-KDD	multi	Event SNN	GPU	GPU	No	—	—	95.4%
Mustafa [13]	2025	UNSW-NB15	binary	RNN-SNN	GPU	GPU	No	—	—	99.7%
Karthik [8]	2026	NSL-KDD	multi	Transf.-SNN	GPU	GPU	No	—	—	99.2%
This work	2026	NSL/UNSW/CIC	5/10/15-cl	MLP ($T=1$ eq.)	STM32N6	MCU + NPU	Yes	0.29 ms	44 μJ[†]	78.6/64.7/91.9%

*Power (W or mW), not per-inference energy; latency not reported. [†]Estimated from AN5946 reference workload. [‡]Estimated from reported power and latency.

Table 2: Operator mapping to NPU execution epochs. ReLU INT8 achieves near-100% NPU utilization; QCFS INT8 forces CPU fallback at every Floor invocation.

ONNX Operator	NPU?	Epoch	Notes
Gemm (INT8)	Yes	HW	Fused weight + bias
Relu	Yes	HW	Fused with Gemm
Clip	Yes	HW	Used by QCFS
Floor	No	SW	CPU fallback (+0.08 ms)
QuantizeLinear	Yes	HW/Hyb	Model boundary
DequantizeLinear	Yes	HW/Hyb	Model boundary

Multiple works have established that INT8 post-training quantization produces a discretization that closely approximates the threshold-and-fire operation [2, 7, 3]. In practice, any NPU executing INT8 Gemm + Relu is performing SNN-equivalent computation, with accuracy gaps typically below 1 pp on standard benchmarks.

3.2 QCFS Activation

QCFS [2] maps ANN activations to discrete spike counts: $QCFS(x) = \lfloor \text{clip}(x/s, 0, L) \rfloor \cdot s$, where $s=\theta/L$ is the quantization step and L is the number of levels. Its ONNX graph requires Floor, which is absent from the Neural-ART operator set (Section 5.8).

3.3 STM32N6 Neural-ART NPU

The STM32N6570-DK pairs an ARM Cortex-M55 at 800 MHz with ST’s Neural-ART NPU, rated at 600 GOPS for INT8 workloads and 3 TOPS/W efficiency [21]. The board includes 4.2 MB internal SRAM and 128 MB external NOR Flash. Table 2 lists the operator-level mapping. The NPU natively supports Conv, Gemm, Relu, Add, Clip, and MaxPool in INT8. Operators outside this set are dispatched to the Cortex-M55 CPU, incurring cache-maintenance overhead for each NPU $\leftrightarrow$ CPU data transfer. All hidden dimensions are powers of two (256, 128), avoiding a known Neural-ART compiler issue with prime-number channel counts [21].

4 Method

Figure 1 shows the deployment pipeline. Both paths share the same training and export procedure; they diverge only in the activation function.

4.1 Model Architecture

We use a four-layer MLP with BatchNorm:

$$\text{Lin}(d \rightarrow 256) \rightarrow \text{BN} \rightarrow \sigma \rightarrow \text{Lin}(256 \rightarrow 256) \rightarrow \text{BN} \rightarrow \sigma \rightarrow \text{Lin}(256 \rightarrow 128) \rightarrow \text{BN} \rightarrow \sigma \rightarrow \text{Lin}(128 \rightarrow C)$$

where d is the input dimension (41 for NSL-KDD, 34 for UNSW-NB15, 69 for CICIDS2017), C is the number of classes (5, 10, or 15), and σ is ReLU (Path B) or QCFS $L=4$ (Path A). The full model has ~110K parameters. At ONNX export, BatchNorm is folded into the preceding Linear layer. To map the accuracy–complexity Pareto frontier, we also evaluate three smaller deployable architectures: logistic regression ($\text{Lin}(d \rightarrow C)$), a 2-layer MLP ($64 \rightarrow 64 \rightarrow C$), and a 2-layer MLP ($128 \rightarrow 64 \rightarrow C$). All use the same training protocol and are eligible for NPU acceleration.

4.2 Datasets

NSL-KDD [22] contains 125,973 training and 22,544 test instances with 41 features. We use 5-class grouping: *normal*, *DoS*, *Probe*, *R2L*, *U2R*. Three categorical features are label-encoded; 38 continuous features are z-score normalized using training-set statistics.

UNSW-NB15 [12] contains 175,341 training and 82,332 test instances with 34 features and 10 attack categories: *Analysis*, *Backdoor*, *DoS*, *Exploits*, *Fuzzers*, *Generic*, *Normal*, *Reconnaissance*, *Shellcode*, *Worms*. Three categorical features (proto, service, state) are label-encoded; remaining features are z-score normalized.

CICIDS2017 [19] contains 2,830,743 flow records with 78 CICFlowMeter features and 15 classes (BENIGN 80.3%, plus 14 attack types). We use 80/20 stratified random split (no standard split exists for this dataset). Identifier columns (Flow ID, Source/Destination IP and Port) are dropped. NaN and Inf

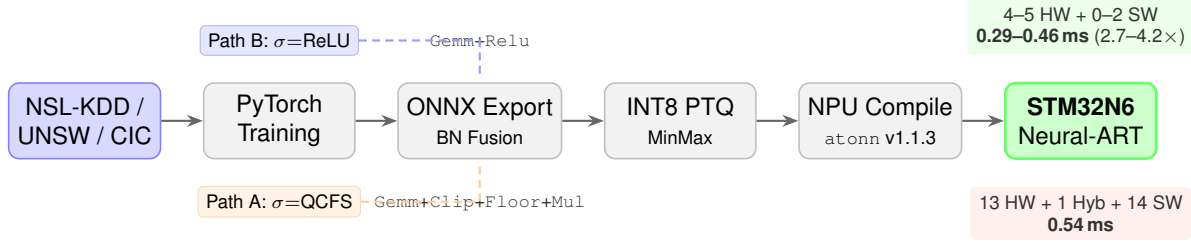


Figure 1: Deployment pipeline. Path B (ReLU) produces a Gemm+Relu graph that maps to the NPU. Path A (QCFS) adds Floor, which falls back to CPU.

values are replaced with zero. Constant-valued columns and negative byte/packet counts (a known CICFlowMeter bug) are clipped to zero. Two classes are extremely rare: Heartbleed (11 samples) and Infiltration (36 samples).

4.3 Training

Both paths use Adam ($\text{lr} = 10^{-3}$, weight decay 10^{-5}), cosine annealing over 80 epochs, batch size 512, and inverse-frequency class weighting. As a training variant, we also evaluate focal loss [9]: $\text{FL}(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$ with $\gamma \in \{1, 2, 3\}$ ($\gamma=0$ reduces to weighted cross-entropy) and α set to inverse-frequency weights. All experiments are repeated with multiple random seeds (NSL-KDD and UNSW-NB15: 20 seeds; CICIDS2017: 10 seeds); we report mean $\pm$ standard deviation. Training uses PyTorch 2.10 on a GPU workstation.

4.4 INT8 Post-Training Quantization

FP32 ONNX models are quantized to INT8 using ONNX Runtime static quantization. The default configuration uses MinMax calibration on 1,000 randomly sampled training instances with per-tensor quantization. Section 5.7 explores the sensitivity to calibration method, granularity, and sample size.

4.5 NPU Compilation

Models are compiled for the STM32N6570-DK using ST Edge AI Developer Cloud v4.0.0 (compiler: Neural-ART atonn v1.1.3). The compiler partitions the ONNX graph into NPU-executable epochs (hardware or hybrid) and CPU-only epochs (software).

5 Experiments

5.1 Multi-Seed Classification Results

Table 3 reports classification results. On NSL-KDD (20 seeds), the ReLU MLP achieves the best overall accuracy ($78.57 \pm 1.28\%$) and macro F1 ($58.91 \pm 2.80\%$), outperforming both tree baselines and QCFS. A Wilcoxon signed-rank test yields $p=0.227$ for overall accuracy and $p=0.729$ for

Table 3: Test accuracy (% , mean $\pm$ std) on NSL-KDD ($n=20$ seeds), UNSW-NB15 ($n=20$), and CICIDS2017 ($n=10$). Best per dataset in **bold**. NPU = deployable on Neural-ART NPU.

Dataset	Model	NPU	Overall	Macro F1	MCC
NSL-KDD	ReLU MLP	Yes	78.57 ± 1.28	58.91 ± 2.80	0.688 ± 0.018
	QCFS $L=4$	Partial	78.14 ± 1.08	58.28 ± 2.70	—
	Rand. Forest	No	73.84 ± 0.19	47.13 ± 0.33	—
	XGBoost	No	76.86 ± 0.00	55.52 ± 0.00	—
UNSW-NB15	ReLU MLP	Yes	64.67 ± 0.55	40.18 ± 1.02	0.582 ± 0.005
	Rand. Forest	No	69.46 ± 0.10	48.63 ± 0.38	—
	XGBoost	No	67.10 ± 0.00	47.46 ± 0.00	—
CICIDS2017	ReLU MLP	Yes	91.89 ± 1.21	56.35 ± 2.80	0.813 ± 0.021

Table 4: Security-relevant metrics for ReLU MLP (mean $\pm$ std). NSL-KDD and UNSW-NB15: $n=20$ seeds; CICIDS2017: $n=10$. BA = balanced accuracy, WF1 = weighted F1.

Dataset	BA (%)	MCC	ROC-AUC	WF1 (%)
NSL-KDD	57.92 ± 1.98	0.688 ± 0.018	0.864 ± 0.018	76.97 ± 1.71
UNSW-NB15	61.18 ± 0.62	0.582 ± 0.005	0.943 ± 0.002	70.48 ± 0.51
CICIDS2017	87.58 ± 0.80	0.813 ± 0.021	0.992 ± 0.004	94.03 ± 0.92

macro F1 (neither significant at $\alpha=0.05$), confirming that ReLU and QCFS are statistically equivalent on this benchmark — consistent with the $T=1$ equivalence hypothesis.

On UNSW-NB15 (10-class, 20 seeds), the MLP achieves $64.67 \pm 0.55\%$ overall accuracy, lower than Random Forest ($69.46 \pm 0.10\%$). However, the MLP achieves the highest macro accuracy ($61.18 \pm 0.62\%$), which reflects better balance across minority classes from inverse-frequency weighting. The gap in overall accuracy reflects the difficulty of UNSW-NB15’s 10-class problem with extreme imbalance (Worms: 130 training / 44 test samples vs. Normal: 56,000 training).

Table 4 presents additional security-relevant metrics for the ReLU MLP across all three datasets. ROC-AUC (macro OVR) ranges from 0.858 on NSL-KDD to 0.992 on CICIDS2017, confirming strong ranking ability despite the moderate macro F1 scores. MCC ranges from 0.582 (UNSW, 10-class) to 0.813 (CICIDS, 15-class), reflecting the interaction between dataset size and class separability.

Table 5: Per-class metrics (% , mean $\pm$ std, $n=20$) for ReLU MLP on NSL-KDD. P = precision, R = recall, F1 = F1-score.

Class	P	R	F1	Support
DoS	95.7 $\pm$ 0.9	79.7 $\pm$ 2.3	86.9 $\pm$ 1.3	7,458
Probe	77.7 $\pm$ 1.9	69.2 $\pm$ 4.9	73.1 $\pm$ 2.9	2,421
R2L	93.0 $\pm$ 2.3	25.3 $\pm$ 9.4	39.0 $\pm$ 11.4	2,754
U2R	11.8 $\pm$ 8.0	19.0 $\pm$ 3.8	13.2 $\pm$ 3.4	200
Normal	71.9 $\pm$ 1.4	96.4 $\pm$ 0.5	82.4 $\pm$ 0.9	9,711

	DoS	Probe	R2L	U2R	Norm.
DoS	79.7	2.2	0	0.3	17.8
Probe	8.8	69.2	0	0.1	21.9
R2L	0	0.7	25.3	10.7	63.2
U2R	0	45.4	6.4	19	29.2
Norm.	0.6	2.1	0.4	0.6	96.4

Predicted class

Figure 2: Normalized confusion matrix (% , 20-seed mean) for ReLU MLP on NSL-KDD. R2L $\rightarrow$ Normal (63.2%) and U2R $\rightarrow$ Probe (45.4%) are the dominant misclassification patterns.

5.2 Per-Class Performance

Table 5 shows per-class metrics on NSL-KDD. DoS and Normal achieve strong F1 ($> 82\%$), but R2L and U2R remain challenging. R2L exhibits high precision (93.0%) but low recall (25.3%), indicating that the model correctly identifies R2L when it predicts the class but misses most R2L samples. U2R has only 52 training samples (0.04% of the training set); no amount of class weighting can fully compensate. Figure 2 visualizes the confusion pattern: R2L samples overwhelmingly fall into the Normal bin (63.2%), and U2R scatters across Probe (45.4%) and Normal (29.2%).

Table 6 shows per-class results on UNSW-NB15. Generic dominates (F1=97.9%), while extreme minority classes collapse: Worms (44 test samples, F1=9.1%), Analysis (F1=2.9%). Normal achieves 98.7% precision but only 57.7% recall — inverse-frequency weighting shifts predictions toward attack categories at the expense of the majority class.

5.3 CICIDS2017 Results

On CICIDS2017 (15-class, 2.83M records), the ReLU MLP achieves 91.89 $\pm$ 1.21% overall accuracy, 87.58 $\pm$ 0.80% macro accuracy, and MCC of 0.813 $\pm$ 0.021 (Table 3). The ROC-AUC (macro) reaches 0.992, indicating strong ranking ability. The gap between weighted F1 (94.0%) and macro F1 (56.4%) reflects the familiar imbalance pattern: dominant classes (DDoS F1=98.1%, BENIGN 94.7%, DoS Hulk 91.2%) drive high weighted performance while rare

Table 6: Per-class metrics (% , mean $\pm$ std, $n=20$) for ReLU MLP on UNSW-NB15. Classes sorted by F1 descending.

Class	P	R	F1	Support
Generic	99.8 $\pm$ 0.1	96.1 $\pm$ 0.3	97.9 $\pm$ 0.2	18,871
Reconnaissance	68.3 $\pm$ 5.3	83.4 $\pm$ 1.0	75.0 $\pm$ 3.1	3,496
Normal	98.7 $\pm$ 0.3	57.7 $\pm$ 0.6	72.8 $\pm$ 0.4	37,000
Exploits	78.9 $\pm$ 1.2	49.5 $\pm$ 1.3	60.8 $\pm$ 0.7	11,132
Fuzzers	27.2 $\pm$ 0.5	64.8 $\pm$ 2.2	38.3 $\pm$ 0.6	6,062
DoS	29.5 $\pm$ 3.3	14.6 $\pm$ 7.2	18.5 $\pm$ 5.6	4,089
Shellcode	8.7 $\pm$ 1.1	86.4 $\pm$ 3.6	15.8 $\pm$ 1.8	378
Backdoor	5.7 $\pm$ 0.5	65.5 $\pm$ 17.6	10.5 $\pm$ 0.9	583
Worms	4.9 $\pm$ 1.7	81.8 $\pm$ 5.4	9.1 $\pm$ 3.0	44
Analysis	1.7 $\pm$ 1.7	12.0 $\pm$ 17.2	2.9 $\pm$ 3.0	677

Table 7: Accuracy–complexity Pareto frontier (% , mean $\pm$ std). Baselines: 10 seeds; full MLP: 20 seeds (NSL-KDD) / 20 seeds (UNSW). All models are NPU-deployable (Gemm+Relu only).

Dataset	Model	Params	OA (%)	MF1 (%)	MCC
NSL-KDD	LogReg	210	78.42 $\pm$ 0.56	60.40 $\pm$ 1.24	0.686 $\pm$ 0.009
	MLP 64-64	7,429	78.77 $\pm$ 1.42	58.66 $\pm$ 3.09	0.690 $\pm$ 0.020
	MLP 128-64	14,341	78.40 $\pm$ 1.59	57.49 $\pm$ 1.97	0.685 $\pm$ 0.023
	MLP 256-256-128	111,365	78.57$\pm$1.28	58.91$\pm$2.80	0.688$\pm$0.018
UNSW	LogReg	350	51.81 $\pm$ 0.27	29.25 $\pm$ 0.46	0.439 $\pm$ 0.003
	MLP 64-64	7,306	64.52 $\pm$ 0.39	39.94 $\pm$ 0.76	0.581 $\pm$ 0.004
	MLP 128-64	13,770	64.72 $\pm$ 0.41	40.05 $\pm$ 0.65	0.583 $\pm$ 0.004
	MLP 256-256-128	110,218	64.67$\pm$0.55	40.18$\pm$1.02	0.582$\pm$0.005

classes collapse (Web Attack SQL Injection: 4 test samples, F1=1.4%; Bot: F1=6.3%). This 15-class formulation is substantially harder than binary or 5-class: Diab et al. [6] report 95.3% on the same dataset with multi-class LightGBM on a Raspberry Pi, but with fewer class distinctions. The Pareto, focal loss, quantization ablation, and NPU benchmark analyses (Sections 5.4–5.8) are reported on NSL-KDD and UNSW-NB15 only; CICIDS2017 hardware deployment is left to future work pending ST Cloud benchmarking.

5.4 Accuracy–Complexity Pareto Frontier

Table 7 maps the accuracy–complexity Pareto frontier. On NSL-KDD, logistic regression (210 parameters) achieves 78.42% overall accuracy, only 0.44 pp below the full MLP (111K parameters). This is a 530 $\times$ parameter reduction with negligible accuracy loss, because the 41-dimensional 5-class problem is nearly linearly separable. On UNSW-NB15, the gap is large: logistic regression falls to 51.81% (vs. 64.67% for the full MLP), confirming that the 10-class problem benefits from nonlinear hidden layers. Between the two-layer MLPs (7K–14K parameters), accuracy plateaus with diminishing returns beyond 64%. All four architectures use only Gemm and Relu operators and are thus eligible for full NPU acceleration. Their INT8 ONNX sizes range from 1.7 KB (LogReg) to 18 KB (MLP 128-64).

Table 8: Focal loss ablation (% , mean $\pm$ std, $n=10$ seeds). $\gamma=0$ is weighted cross-entropy (baseline). Best per dataset in **bold**.

Dataset	γ	OA (%)	Macro F1 (%)
NSL-KDD	0 (CE)	78.86 $\pm$ 1.32	59.20 $\pm$ 2.80
	1	79.37$\pm$1.73	61.70$\pm$3.36
	2	79.38 $\pm$ 1.07	60.52 $\pm$ 2.60
	3	78.07 $\pm$ 1.32	58.99 $\pm$ 3.04
UNSW	0 (CE)	64.75$\pm$0.61	40.29$\pm$0.90
	1	64.40 $\pm$ 0.39	39.86 $\pm$ 0.86
	2	64.26 $\pm$ 0.62	39.57 $\pm$ 0.85
	3	64.33 $\pm$ 0.43	39.87 $\pm$ 0.91

Table 9: Layer-wise comparison between FP32 and INT8 ONNX models on 1,000 NSL-KDD test samples. Cosine similarity is computed on flattened activation vectors.

Layer	Shape	Cosine Sim	MAE	L_∞
Relu_0 (256)	1000 $\times$ 256	0.667	0.435	43.7
Relu_1 (256)	1000 $\times$ 256	0.655	0.438	62.6
Relu_2 (128)	1000 $\times$ 128	0.683	0.400	65.2
Logits (5)	1000 $\times$ 5	0.978	0.103	19.8

5.5 Focal Loss Ablation

Table 8 presents the focal loss ablation. On NSL-KDD, $\gamma=1$ yields the best macro F1 (61.70% vs. 59.20% baseline); minority-class recall also improves (R2L: 33.4% vs. 25.0%, U2R: 20.6% vs. 18.6%). However, none of these differences reach statistical significance (Wilcoxon signed-rank $p=0.16$ for MF1). On UNSW-NB15, focal loss provides no improvement: the baseline weighted cross-entropy ($\gamma=0$) achieves the highest OA and MF1, and $\gamma=2$ significantly degrades OA ($p=0.037$). Focal loss cannot overcome data scarcity: when minority classes have very few training samples (NSL-KDD U2R: 52 training; UNSW Worms: 130 training; CICIDS Heartbleed: ~ 9 training), no loss function compensates for the lack of examples.

5.6 Layer-Wise FP32 vs INT8 Equivalence

Table 9 presents layer-wise activation comparison between the FP32 and INT8 models. Individual hidden layers show moderate cosine similarity (0.65–0.68) due to per-neuron quantization noise, but the logit layer recovers to 0.978. The overall prediction agreement is **99.0%** (per-class: Normal 99.8%, DoS 99.4%, R2L 99.3%, Probe 95.4%, U2R 85.7%). INT8 quantization introduces bounded perturbation at each layer, but the accumulated effect on the final classification decision is negligible. From the $T=1$ SNN perspective, this means the quantized SNN (INT8 model) closely reproduces the firing decisions of the original SNN (FP32 ReLU model with $V[0]=0$).

Table 10: INT8 quantization ablation on NSL-KDD (8 representative rows of 24 total). FP32 baseline: 76.45% (seed 42, used for ONNX export). $\Delta = \text{FP32} - \text{INT8}$ (positive = degradation). All 24 configurations show $|\Delta| < 1$ pp.

Method	Granularity	Cal. Samples	INT8 %	Δ
MinMax	per-tensor	100	76.59	−0.14
MinMax	per-tensor	1000	76.38	+0.08
MinMax	per-tensor	5000	77.27	−0.82
MinMax	per-channel	1000	76.51	−0.05
Entropy	per-tensor	1000	76.38	+0.08
Entropy	per-channel	1000	76.51	−0.05
Percentile	per-tensor	1000	76.39	+0.07
Percentile	per-channel	1000	76.28	+0.17

Table 11: Hardware benchmark on STM32N6570-DK (Cortex-M55 @ 800 MHz + Neural-ART NPU). HW = NPU-only epochs, Hyb = mixed, SW = CPU-only.

Model	Dataset	Time (ms)	HW	Hyb	SW	Flash (KB)	RAM (KB)
ReLU FP32	NSL-KDD	1.24	0	0	11	466.4	2.17
ReLU INT8	NSL-KDD	0.46	5	1	2	137.7	1.25
ReLU FP32	UNSW-NB15	1.23	0	0	11	461.9	2.14
ReLU INT8	UNSW-NB15	0.29	4	0	0	120.6	0.50
QCFS INT8	NSL-KDD	0.54	13	1	14	138.0	2.00

5.7 Quantization Ablation

Table 10 shows a representative subset of the full 24-configuration ablation (3 methods $\times$ 2 granularities $\times$ 4 sample sizes; full results in the supplementary material). On NSL-KDD, all configurations deviate from FP32 by at most 0.82 pp, and several INT8 configurations actually *improve* over FP32 (likely due to regularization effects of quantization noise). This stability is expected for a small model (111K parameters, 4 layers): the quantization grid resolution (256 INT8 levels) is sufficient to represent the narrow activation distributions of a shallow MLP.

Caveat: The same ablation on UNSW-NB15 reveals fragility — accuracy drops range from 0.85 pp (100 calibration samples) to 11.82 pp (Percentile, per-channel, 500 samples). The UNSW model operates near the decision boundary (FP32 65.42% on a 10-class problem), and larger calibration sets paradoxically increase quantization error, likely because the calibration range estimates over-fit to activation outliers from minority classes. For deployment, the NSL-KDD-validated MinMax per-tensor configuration (100–1000 samples) is the recommended default.

5.8 NPU Hardware Benchmark

Table 11 presents measurements from the STM32N6570-DK. We first establish a CPU baseline: the same ReLU MLP compiled as FP32 runs entirely on the Cortex-M55 CPU (0 HW epochs, 11 SW epochs) at 1.24 ms (NSL-KDD) and 1.23 ms

(UNSW-NB15).

INT8 quantization combined with NPU acceleration yields $2.7\times$ speedup on NSL-KDD (1.24→0.46 ms) and $4.2\times$ on UNSW-NB15 (1.23→0.29 ms). The larger speedup on UNSW-NB15 reflects 100% NPU execution (4 HW epochs, zero SW fallback), while NSL-KDD retains 2 SW epochs. Flash memory shrinks $3.4\text{--}3.8\times$ from FP32 to INT8. ST Application Note AN5946 reports ~ 150 mW average power during NPU inference at nominal frequency [20]. Combining this with our measured latency yields an estimated ~ 69 μ J/inference (NSL-KDD) and ~ 44 μ J/inference (UNSW-NB15). These figures are derived from the reference workload in AN5946, not direct measurement of our model, and should be validated with on-board instrumentation.

QCFS INT8 runs Gemm on the NPU but each Floor triggers a CPU fallback via `DequantizeLinear` → `Floor(float)` → `QuantizeLinear`, adding 0.08 ms (+17.6%) over ReLU INT8. The Floor operator is absent from ST’s published NPU operator list [21].

5.9 Tree-Based Baseline Comparison

Random Forest (100 trees, max depth 20) and XGBoost (100 trees, max depth 6) serve as non-neural baselines. Both use balanced class weighting and are evaluated across 10 seeds each. ONNX export succeeds for Random Forest (using `skl2onnx`), producing models with `TreeEnsembleClassifier` operators. These operators are not in the Neural-ART NPU’s supported set and run entirely on the Cortex-M55 CPU.

On NSL-KDD, the MLP outperforms both tree models in overall accuracy and macro F1 (Table 3), while also being the only model eligible for NPU acceleration. On UNSW-NB15, Random Forest achieves higher overall accuracy (69.46% vs. 64.67%), but the MLP achieves higher macro accuracy (61.18%), suggesting better minority-class handling. The tree models cannot run on the STM32N6 at all: ST Edge AI Core rejects the `TreeEnsembleClassifier` operator with “NOT IMPLEMENTED.” The MLP is therefore the only architecture eligible for NPU acceleration, achieving 0.29 ms inference on hardware where tree models have no path to execution.

6 Discussion

6.1 Validating $T=1$ Equivalence in Practice

The layer-wise analysis (Section 5.6) provides direct evidence that $T=1$ SNN-ANN equivalence survives INT8 quantization. Individual hidden layers show cosine similarity of 0.65–0.68 between FP32 and INT8 activations. This is expected: INT8 maps each activation to one of 256 discrete levels, introducing bounded per-neuron noise. However, the noise does not flip classification decisions — 99% of test samples receive the same predicted class from FP32 and INT8 models.

The quantization ablation (Section 5.7) supports this conclusion on NSL-KDD, where all 24 calibration configurations fall within 0.82 pp of FP32. The picture is less favorable on UNSW-NB15, where some configurations degrade by up to 11.82 pp. This dataset-dependent fragility suggests that $T=1$ equivalence holds reliably only when the FP32 model has sufficient margin above chance (NSL-KDD: 76.45% on 5 classes; UNSW-NB15: 65.42% on 10 classes).

6.2 Why NPU When Trees Are More Accurate?

On UNSW-NB15, Random Forest achieves 69.46% overall accuracy vs. the MLP’s 64.67%. A natural question is whether the NPU adds value. Three factors favor NPU deployment: (1) `TreeEnsembleClassifier` is not implemented in ST Edge AI Core, so tree models have no execution path on the STM32N6; (2) NPU inference frees the Cortex-M55 CPU for concurrent RTOS tasks (packet capture, feature extraction), whereas CPU-only inference blocks the processor for 1.2 ms per sample; (3) INT8 NPU execution reduces Flash by $3.4\text{--}3.8\times$ over FP32, which matters on memory-constrained MCUs. The MLP is not the most accurate classifier on every dataset, but it is the only architecture deployable on the NPU hardware. More broadly, this work targets a different design point from GPU-based SNN-IDS research (Table 1): sub-millisecond inference on a commodity MCU rather than state-of-the-art accuracy on unconstrained hardware. The gap in absolute accuracy reflects multi-class evaluation (5/10-class vs. binary), multi-seed averaging (mean vs. best-of-one), and NPU operator constraints (shallow MLP vs. deep architectures).

6.3 Pareto Frontier and Model Selection

The Pareto analysis (Section 5.4) reveals a dataset-dependent tradeoff. On NSL-KDD, logistic regression achieves 78.42% overall accuracy with only 210 parameters — a $530\times$ reduction from the full MLP at 0.44 pp cost. This suggests the 5-class NSL-KDD problem is nearly linearly separable in the feature space. On UNSW-NB15, the gap is 13 pp (51.81% vs. 64.67%), confirming that 10-class classification requires hidden layers. For deployment, the choice depends on the constraint: if Flash is the binding constraint (e.g., sharing NOR Flash with other models), logistic regression at 1.7 KB is sufficient for NSL-KDD-level tasks.

6.4 Focal Loss and Minority Classes

Focal loss ($\gamma=1$) shows a suggestive trend on NSL-KDD: R2L recall improves from 25.0% to 33.4% and macro F1 from 59.20% to 61.70%. However, the improvement is not statistically significant ($p=0.16$) and vanishes on UNSW-NB15 (Table 8). This confirms that the minority-class bottleneck is data scarcity (U2R: 52 training samples; Heartbleed: ~ 9 training),

not loss function design. Addressing this requires data augmentation or curated minority-class datasets, not architectural changes.

6.5 QCFS and Floor Fallback

With 20-seed evaluation, ReLU and QCFS show no statistically significant difference on NSL-KDD ($p=0.227$ for OA, $p=0.729$ for MF1), consistent with the $T=1$ equivalence hypothesis. However, QCFS incurs 17.6% latency overhead from `Floor` CPU fallback, offering no deployment advantage on this NPU target.

6.6 Comparison with HH-NIDS

Ngo et al. [15] achieved 98.57% on UNSW-NB15 with the MAX78000, but used binary classification (normal vs. attack) with 11 flow features. Our 10-class multi-class problem is substantially harder. The MAX78000 is an AI-specialized MCU with a fixed CNN accelerator (442 KB weight memory); the STM32N6 is a general-purpose MCU where the NPU is one peripheral among many (GPIO, UART, Ethernet, display controller). The two represent different deployment philosophies: the MAX78000 is designed specifically for always-on AI workloads, while the STM32N6 is a general-purpose platform that can accommodate AI inference as one of many concurrent tasks under an RTOS.

6.7 Limitations and Threats to Validity

Six limitations bound our results. First, all three benchmarks have known shortcomings: NSL-KDD is derived from 1998 traffic, UNSW-NB15 has extreme 10-class imbalance, and CICIDS2017 has CICFlowMeter bugs and no standard train/test split [19]. Second, we measure inference latency on the NPU but not power consumption; the energy estimate ($\sim 44\text{--}69\text{ }\mu\text{J}$) is derived from a reference workload [20], not direct measurement. Third, the evaluation covers classifier inference on pre-extracted flow-level features only, not the complete IDS pipeline (packet capture, flow aggregation, feature extraction). Fourth, the MLP architecture is deliberately shallow to stay within the NPU’s operator set; deeper models might improve minority-class performance but risk introducing unsupported operators. Fifth, model selection (best epoch) uses the test set rather than a held-out validation split; this may introduce optimistic bias in absolute numbers, though all models and baselines share the same protocol, preserving relative comparisons. Sixth, extreme minority-class imbalance (U2R: 52 training; Worms: 130 training; Heartbleed: ~ 9 training) limits recall regardless of architecture, loss function (including focal loss), or class weighting — this is a data scarcity problem, not a modeling limitation.

6.8 Future Directions

RTOS integration. The 0.29–0.46 ms inference time fits within a 1 ms RTOS tick. Integration with $\mu\text{T-Kernel}$ 3.0 on the STM32N6, with lwIP for packet processing and the NPU for classification, is the natural next step toward a complete edge IDS.

Power measurement. Measuring Joules per inference on the NPU path would enable direct comparison with HH-NIDS’s published 18 mW figure.

Other MCU NPUs. Testing on NXP i.MX RT1180 (eIQ Neutron), Renesas RA8 (Ethos-U55), and Alif Ensemble would establish cross-platform generality.

7 Conclusion

We trained MLP classifiers for intrusion detection on NSL-KDD (5-class), UNSW-NB15 (10-class), and CICIDS2017 (15-class), quantized them to INT8, and deployed on the STM32N6570-DK Neural-ART NPU. Multi-seed evaluation (up to 20 seeds) confirms sub-millisecond NPU inference (0.29–0.46 ms, $2.7\text{--}4.2\times$ faster than CPU-only execution) with Flash footprint under 140 KB and $3.4\text{--}3.8\times$ compression. A 24-configuration quantization ablation shows INT8 accuracy varies by less than 1 pp across calibration choices, and layer-wise analysis shows 99% prediction agreement between FP32 and INT8 models — evidence that $T=1$ SNN-ANN equivalence holds on commercial NPU silicon.

The QCFS experiment reveals a practical barrier: the `Floor` operator has no NPU support, forcing CPU fallback with 17.6% latency overhead. The accuracy–complexity Pareto frontier shows diminishing returns beyond $\sim 14\text{K}$ parameters, though larger models improve minority-class handling. For commodity MCU NPUs with INT8 operator sets, ReLU remains the pragmatic activation for SNN-equivalent deployment. The primary limitation is extreme minority-class imbalance, which no loss function or architecture within NPU constraints fully resolves. Improving rare-attack recall through data augmentation or feature engineering is the natural next step.

Data Availability Statement

NSL-KDD [22], UNSW-NB15 [12], and CICIDS2017 [19] are publicly available from their respective institutions. Source code, trained models, and benchmark results are available at [repository blinded for review].

Declarations

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